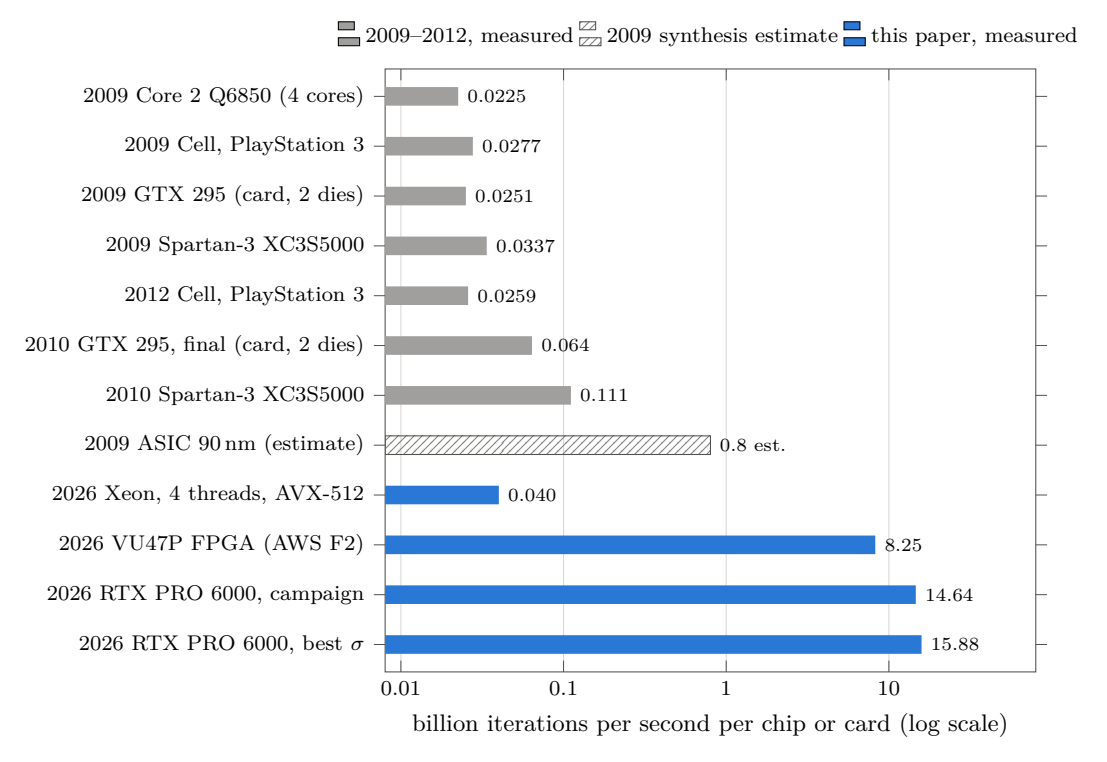
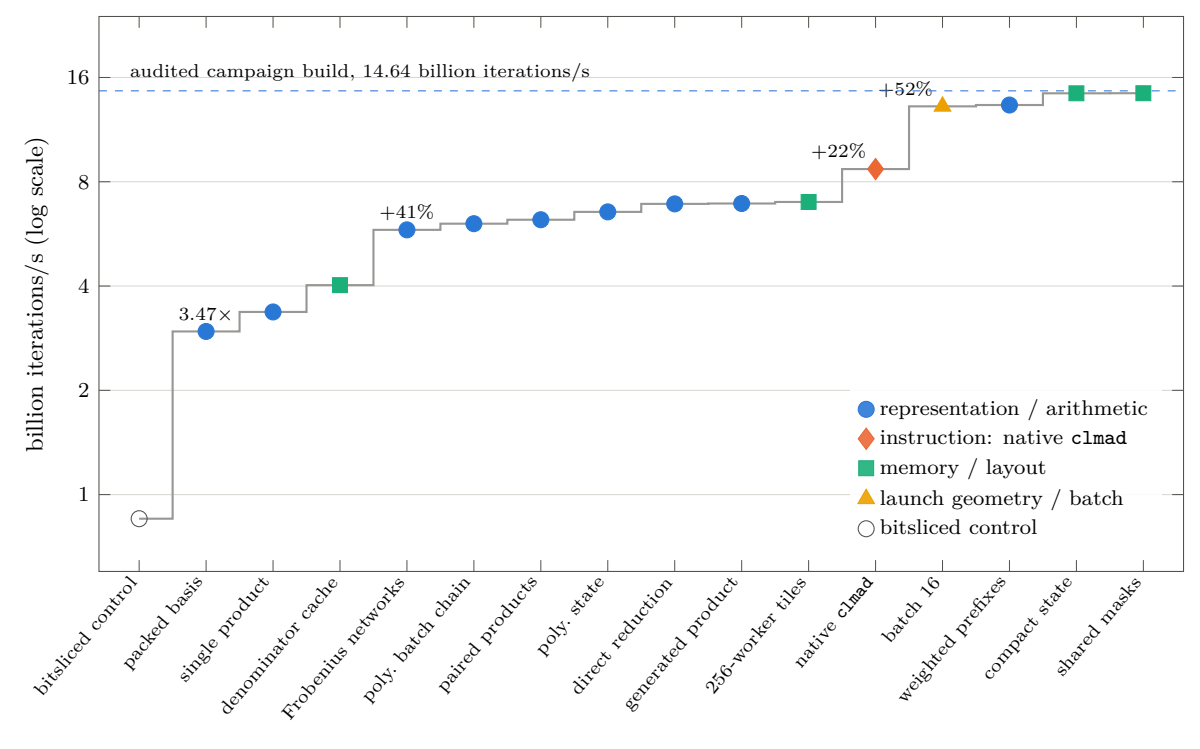
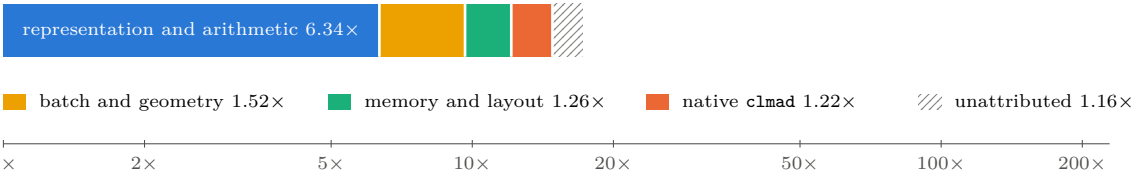

2009–2012, measured
2009 synthesis estimate
this paper, measured
2009 Core 2 Q6850 (4 cores)
0.0225
2009 Cell, PlayStation 3
0.0277
2009 GTX 295 (card, 2 dies)
0.0251
2009 Spartan-3 XC3S5000
0.0337
2012 Cell, PlayStation 3
0.0259
2010 GTX 295, final (card, 2 dies)
0.064
2010 Spartan-3 XC3S5000
0.111
2009 ASIC 90 nm (estimate)
0.8 est.
2026 Xeon, 4 threads, AVX-512
0.040
2026 VU47P FPGA (AWS F2)
8.25
2026 RTX PRO 6000, campaign
14.64
2026 RTX PRO 6000, best σ
15.88
0.01
0.1
1
10
billion iterations per second per chip or card (log scale)


audited campaign build, 14.64 billion iterations/s
billion iterations/s (log scale)
16
8
4
2
1
3.47×
+41%
+22%
+52%
representation / arithmetic
instruction: native clmad
memory / layout
launch geometry / batch
bitsliced control
bitsliced control
packed basis
single product
denominator cache
Frobenius networks
poly. batch chain
paired products
poly. state
direct reduction
generated product
256-worker tiles
native clmad
batch 16
weighted prefixes
compact state
shared masks

Against the 2010 GTX 295 card: 229×



The ledger, bitsliced control to audited build: 17.2×



Grid: 1,504 blocks of 256 threads, run as four waves

wave 1: 376 blocks	wave 2: 376 blocks	wave 3: 376 blocks	wave 4: 376 blocks

385,024 workers = 4× the 96,256 that fit at once (188 SMs × 2 blocks × 256 threads)

One SM (of 188)

block 0: 256 threads

block 1: 256 threads

shared: 1,792-byte σ mask table per block

16 warps per SM; 104 registers per thread

One thread = one worker = 16 walks

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15

walks 0–15: one field inversion per step serves all 16 (Montgomery's trick)

Where the state lives

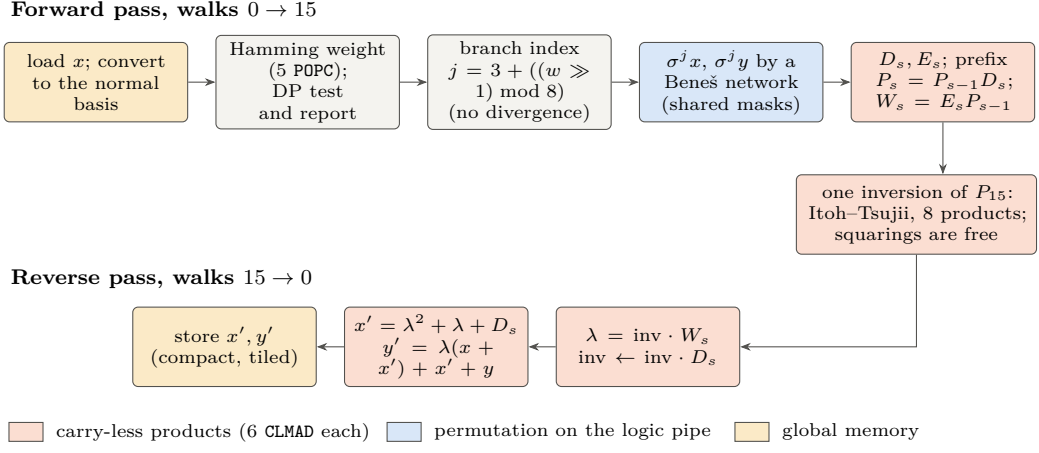
registers: running product and inverse; the current walk's x, y, D, E, λ; unreduced products

shared / constant: σ masks (1,792 B); inverse-network masks; start-point tables

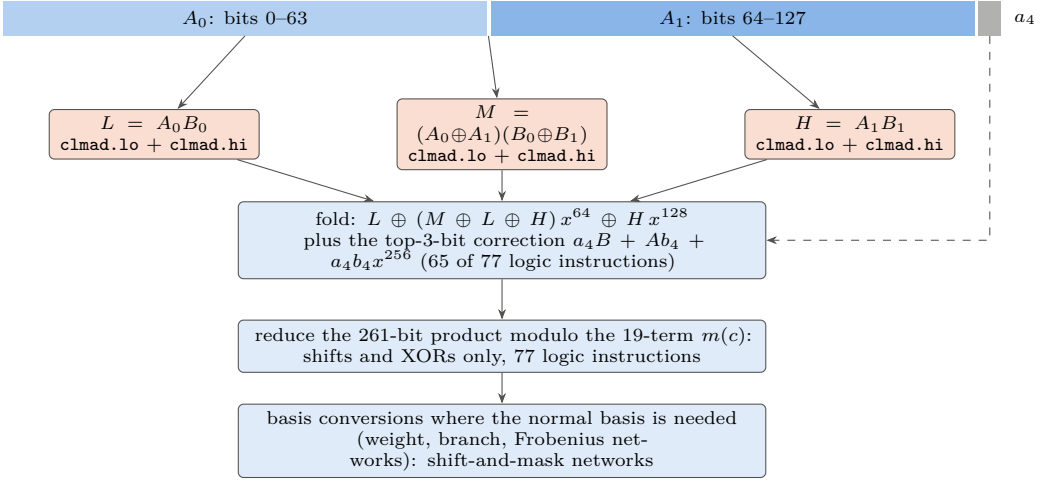
global: x, y, prefix products and cached denominators, 17 B per element in 256-worker tiles

distinguished points: `atomicAdd` into a device buffer, copied to the host between launches

checkpoint: 60 bytes per walk, about 370 MB for the whole grid


Forward pass, walks $0 \to 15$
load x; convert to the normal basis
Hamming weight (5 POPC); DP test and report
branch index $j = 3 + ((w \gg 1) \bmod 8)$ (no divergence)
$\sigma^j x$, $\sigma^j y$ by a Beneš network (shared masks)
D_s, E_s; prefix $P_s = P_{s-1} D_s$; $W_s = E_s P_{s-1}$
one inversion of P_{15}: Itoh–Tsujii, 8 products; squarings are free
Reverse pass, walks $15 \to 0$
$\lambda = \text{inv} \cdot W_s$ $\text{inv} \leftarrow \text{inv} \cdot D_s$
$x' = \lambda^2 + \lambda + D_s$ $y' = \lambda(x + x') + x' + y$
store x', y' (compact, tiled)
carry-less products (6 CLMAD each)
permutation on the logic pipe
global memory

A 131-bit operand in five 32-bit words, read as two 64-bit limbs and 3 top bits



Untiled: one array per (slot, word) spanning all 385,024 workers



a block's 256 threads touch 80 regions, each 1.5 MB apart

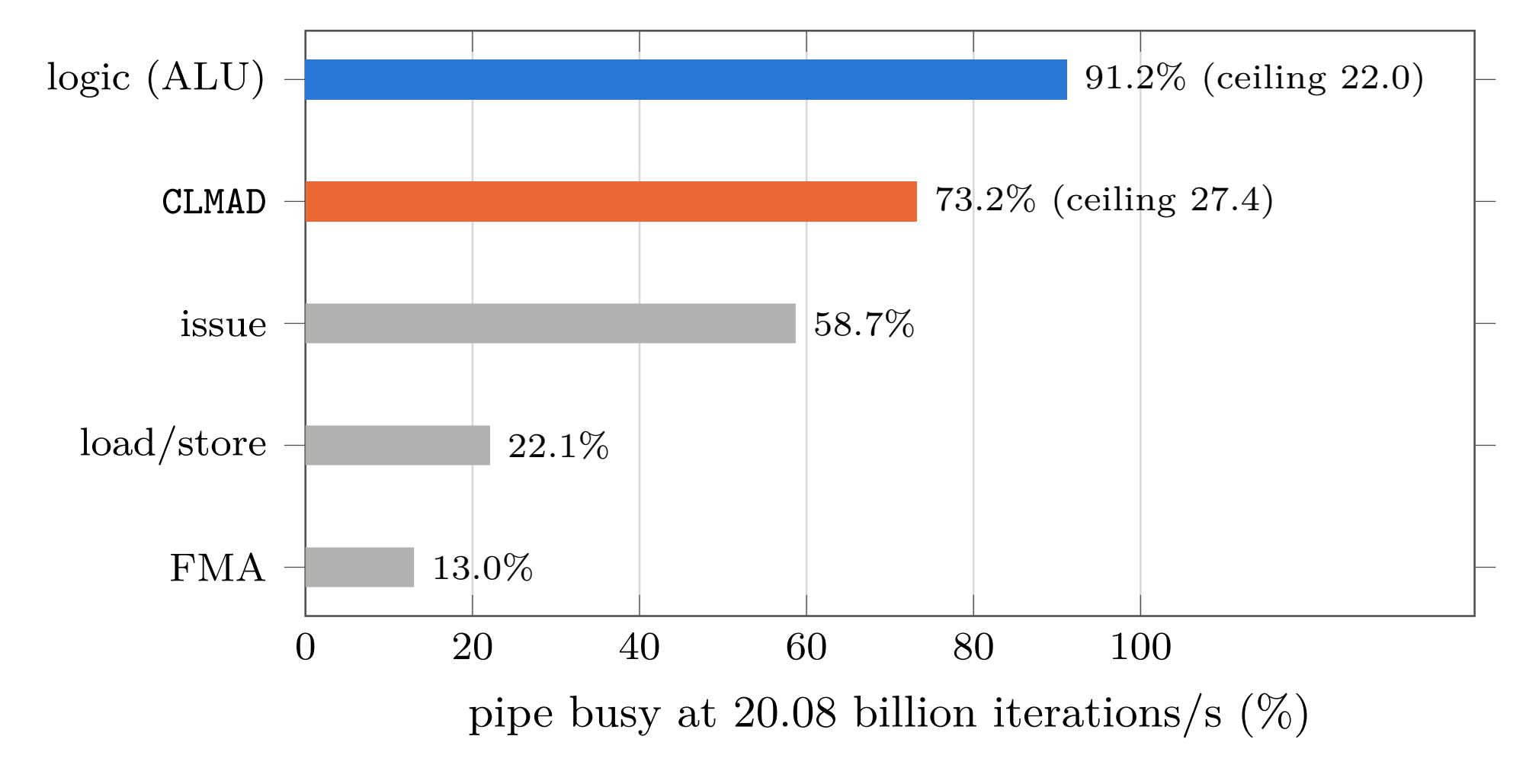
Tiled: each block's 256 workers keep all their slots and words together



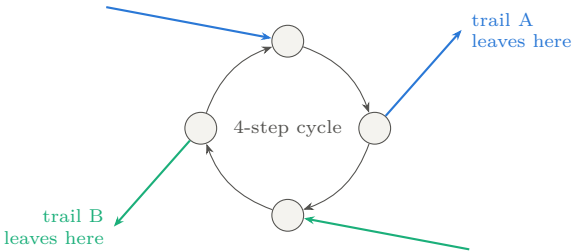
Compact element record for one (tile, slot): 4,352 bytes



17 bytes per element instead of 20: the top 3 bits (or 6, with the cached branch) share one byte; the four state arrays fall from 470 to 399.5 MiB.

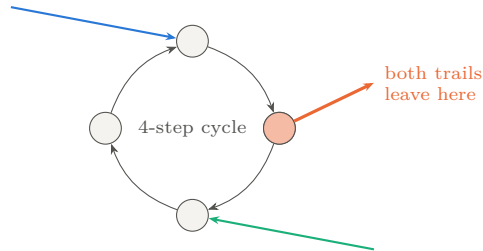

logic (ALU)
CLMAD
issue
load/store
FMA
91.2% (ceiling 22.0)
73.2% (ceiling 27.4)
58.7%
22.1%
13.0%
0
20
40
60
80
100
pipe busy at 20.08 billion iterations/s (%)

Rule v2: the exit depends on the history

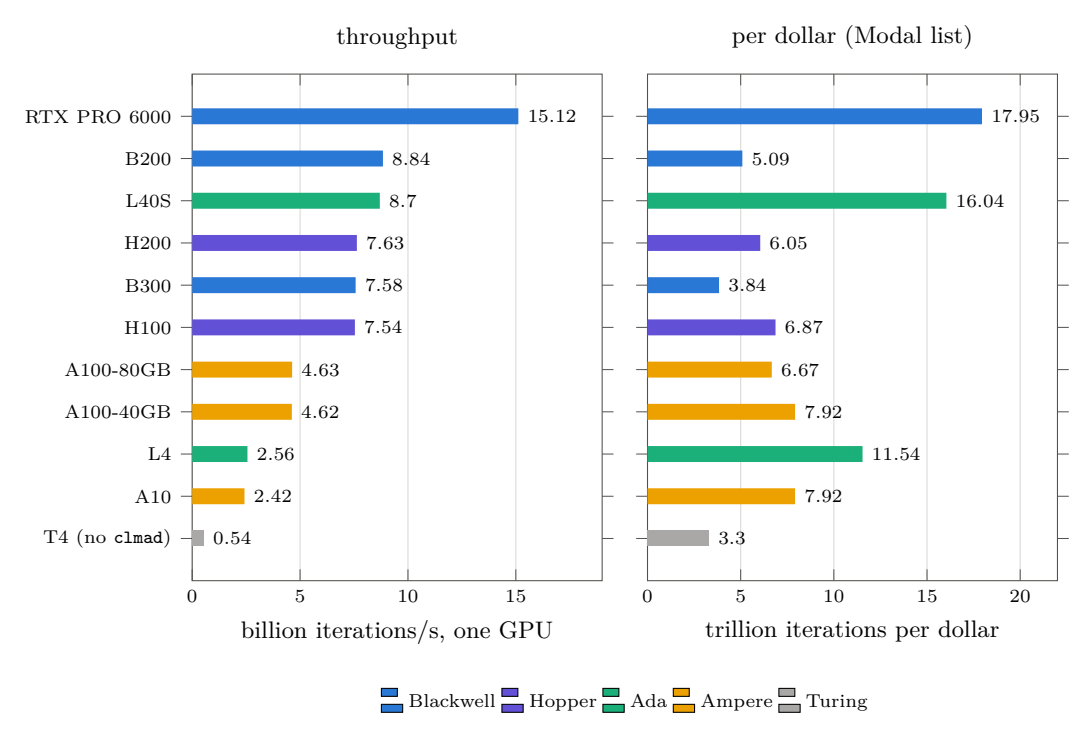


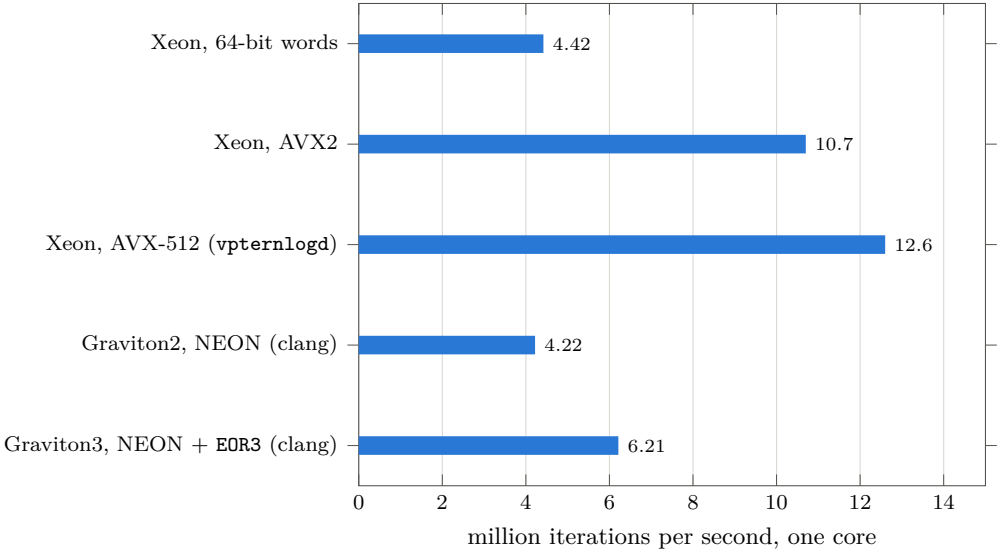
the two trails need not merge,
so their collision can be missed

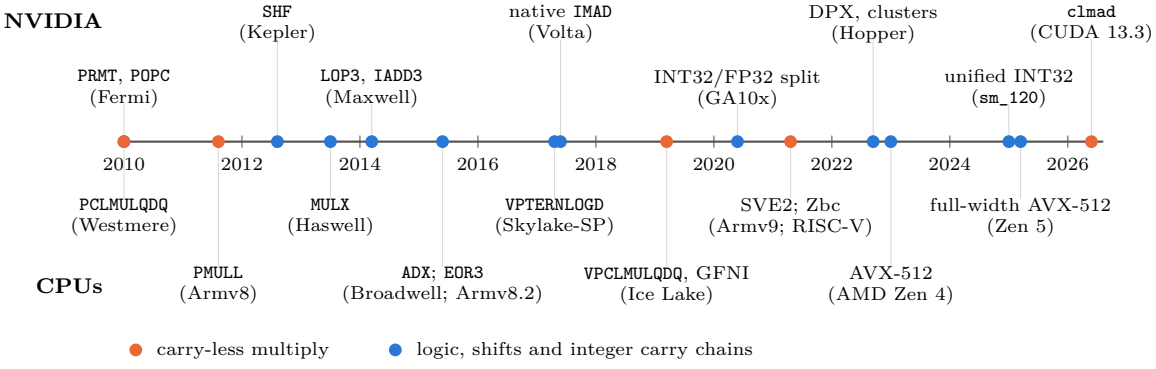
Rule v3: one anchor per cycle

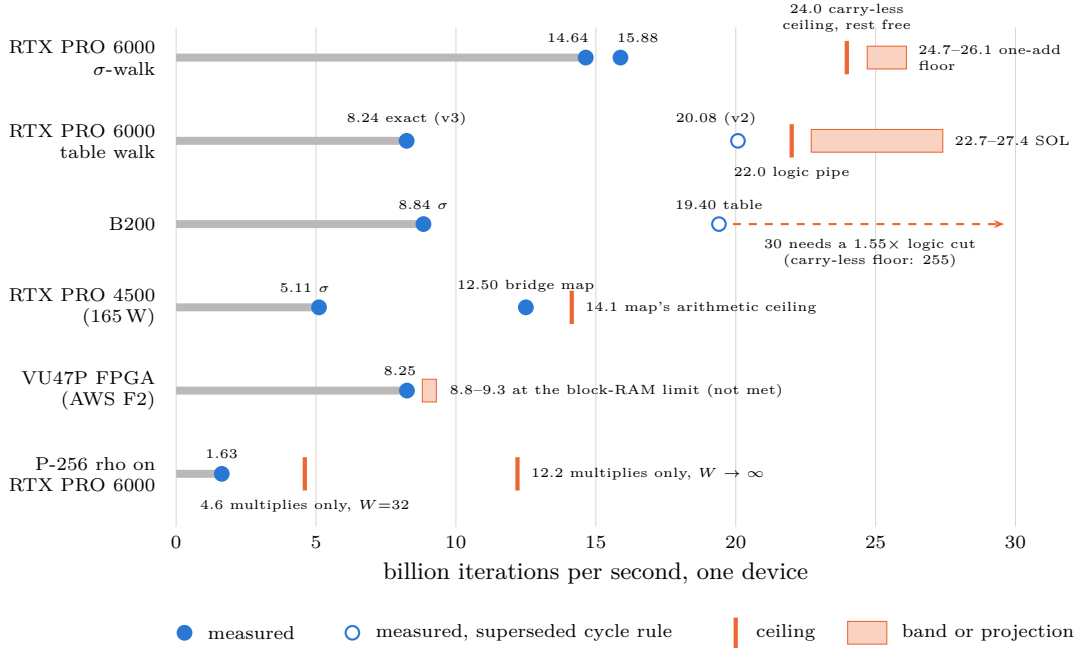


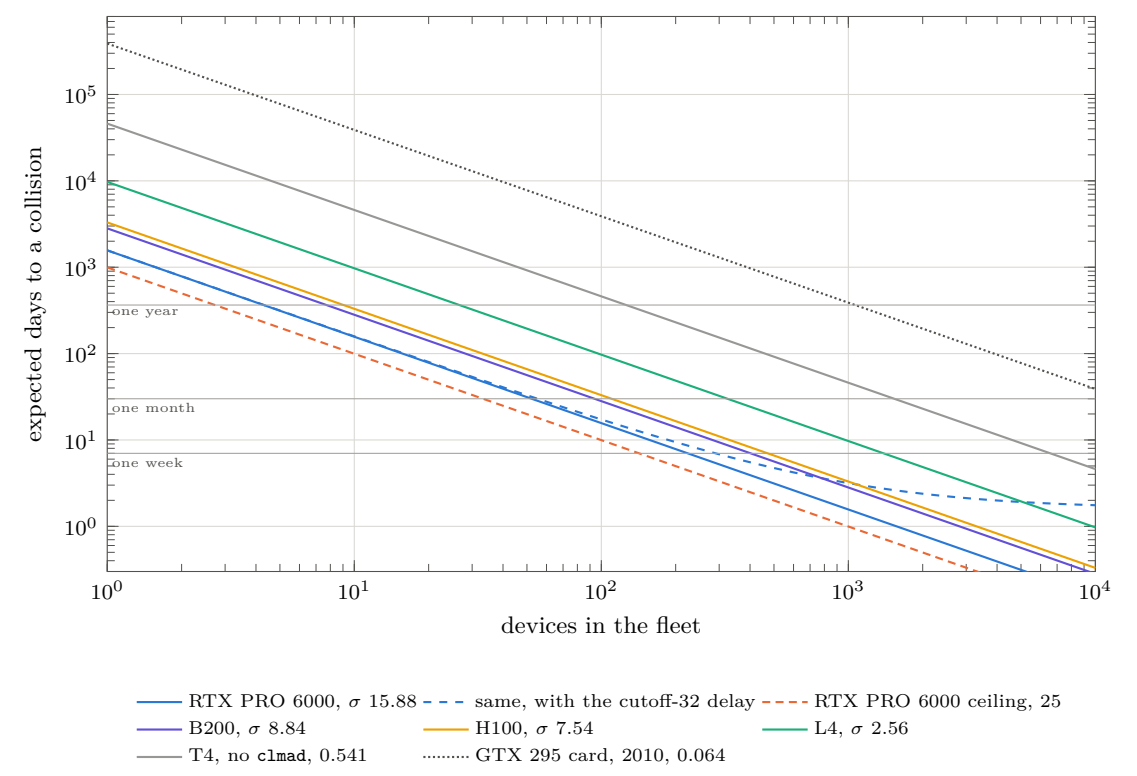
exit chosen from the cycle itself
(least weight and x), so trails coalesce


throughput
per dollar (Modal list)
RTX PRO 6000
B200
L40S
H200
B300
H100
A100-80GB
A100-40GB
L4
A10
T4 (no clmad)
15.12
8.84
8.7
7.63
7.58
7.54
4.63
4.62
2.56
2.42
0.54
17.95
5.09
16.04
6.05
3.84
6.87
6.67
7.92
11.54
7.92
3.3
0
5
10
15
20
billion iterations/s, one GPU
trillion iterations per dollar
Blackwell
Hopper
Ada
Ampere
Turing


Xeon, 64-bit words
4.42
Xeon, AVX2
10.7
Xeon, AVX-512 (vpternlogd)
12.6
Graviton2, NEON (clang)
4.22
Graviton3, NEON + EOR3 (clang)
6.21
0
2
4
6
8
10
12
14
million iterations per second, one core


NVIDIA
SHF
(Kepler)
native IMAD
(Volta)
DPX, clusters
(Hopper)
clmad
(CUDA 13.3)
PRMT, POPC
(Fermi)
LOP3, IADD3
(Maxwell)
INT32/FP32 split
(GA10x)
unified INT32
(sm_120)
2010
2012
2014
2016
2018
2020
2022
2024
2026
PCLMULQDQ
(Westmere)
MULX
(Haswell)
VPTERNLOGD
(Skylake-SP)
SVE2; Zbc
(Armv9; RISC-V)
full-width AVX-512
(Zen 5)
CPUs
PMULL
(Armv8)
ADX; EOR3
(Broadwell; Armv8.2)
VPCLMULQDQ, GFNI
(Ice Lake)
AVX-512
(AMD Zen 4)
carry-less multiply
logic, shifts and integer carry chains


RTX PRO 6000
σ-walk
14.64
15.88
24.0 carry-less
ceiling, rest free
24.7–26.1 one-add
floor
RTX PRO 6000
table walk
8.24 exact (v3)
20.08 (v2)
22.7–27.4 SOL
22.0 logic pipe
B200
8.84 σ
19.40 table
30 needs a 1.55× logic cut
(carry-less floor: 255)
RTX PRO 4500
(165 W)
5.11 σ
12.50 bridge map
14.1 map's arithmetic ceiling
VU47P FPGA
(AWS F2)
8.25
8.8–9.3 at the block-RAM limit (not met)
P-256 rho on
RTX PRO 6000
1.63
4.6 multiplies only, W=32
12.2 multiplies only, W → ∞
0
5
10
15
20
25
30
billion iterations per second, one device
measured
measured, superseded cycle rule
ceiling
band or projection


expected days to a collision
devices in the fleet
one year
one month
one week
RTX PRO 6000, σ 15.88
same, with the cutoff-32 delay
RTX PRO 6000 ceiling, 25
B200, σ 8.84
H100, σ 7.54
L4, σ 2.56
T4, no clmad, 0.541
GTX 295 card, 2010, 0.064